19 Viruses

KEY CONCEPTS

Study Tip

Make flowcharts: Make a simple flowchart for each viral replicative cycle in this chapter. Below is the beginning of an example for the lysogenic cycle. Note the similarities and differences between the replicative cycles.

Lysogenic cycle

Virus attaches to bacterial cell and injects genome.

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Viral genome is integrated into the bacterial chromosome.

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Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 19
- HHMI Animation: HIV Replicative Cycle
- HHMI Video: Interview with Katie Walter

For Instructors to Assign (in Item Library)
- Tutorial: Viral Replication
- Activity: Phage Lysogenic and Lytic Cycles
- Tutorial: CRISPR: The Discovery of Bacterial Adaptive Immunity

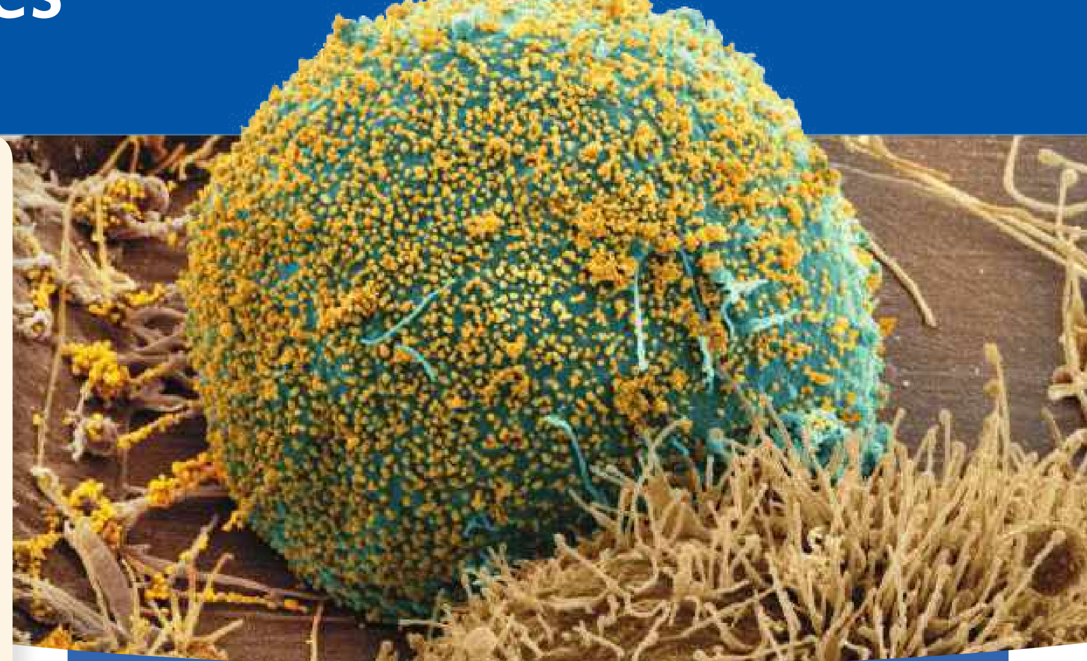

Figure 19.1 A human immune cell (blue) infected by human immunodeficiency virus (HIV) is releasing more HIV viruses (gold dots), which will go on to infect other cells. Left untreated, HIV destroys vital immune system cells, causing AIDS. This disease has killed about 35 million people worldwide.

How does a virus make more viruses?

A virus consists only of nucleic acid, proteins, and sometimes a membranous envelope. After infecting a host cell, it uses the host cell's molecules to make new viruses, as shown in this simplified diagram.